

PHYS 143 – Mechanics and Thermal Physics

Potential Energy and Energy Conservation

What happens to energy when forces act along a path, and how can stored energy predict motion?

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Reading: Young and Freedman, *University Physics* (15th ed.), Ch. 7, §7.1–7.5

By the end of this week, you should be able to:

1. Calculate gravitational and elastic potential energy, and say when a force has a potential energy.
2. Apply conservation of mechanical energy to find speeds and heights.
3. Include friction and other forces in an energy equation, and say where the energy goes.
4. Find a force from a potential energy, and read an energy diagram.

The detailed outcomes are in the checklist at the end of the lecture.

Three contacts:

1 IDENTIFY

What is potential energy, and when is mechanical energy conserved?

2 QUANTIFY

How do springs, friction and other forces fit in?

3 EXTEND

Which forces have a potential energy, and what does $U(x)$ tell us?

Where we start: a five-minute check

Not marked. Answer individually, then we review.

1. A 2.0 kg ball falls 5.0 m. Find the work done by gravity.
2. A 3.0 kg block moving at 4.0 m/s slows to 2.0 m/s. Find the net work done on it.
3. A spring ($k = 200 \text{ N/m}$) is stretched 0.10 m. Find the work done on the spring.
4. A box slides 4.0 m down a rough slope against a 12 N friction force. Find the work done by friction.
5. A 0.50 kg stone moves in a circle of radius 1.0 m at 4.0 m/s. Find its acceleration and the net force.
6. Differentiate $y = 3x^2 - x^3$ and evaluate the derivative at $x = 2$.

1. $W = mgh = (2.0)(9.8)(5.0) = 98 \text{ J}$
(positive: down along the force).
2. $W = \Delta K = \frac{1}{2}(3.0)(2.0^2 - 4.0^2) = -18 \text{ J}.$
3. $W = \frac{1}{2}kx^2 = \frac{1}{2}(200)(0.10)^2 = 1.0 \text{ J}.$
4. $W_f = -fd = -(12)(4.0) = -48 \text{ J}.$
5. $a = v^2/r = 16 \text{ m/s}^2$ toward the centre;
 $F = ma = 8.0 \text{ N}.$
6. $y' = 6x - 3x^2$; at $x = 2$: $12 - 12 = 0.$

Week 2 (free fall), Week 3 (circular motion, friction) and Week 4 (work, the work–energy theorem, springs) are all reused this week. A short derivative refresher is in the supplementary notes.

Contact 1: Identify

*What is potential energy, and when
is mechanical energy conserved?*

Week 4

$$W_{\text{tot}} = \Delta K$$

How does the work done on an object change its speed?

Week 5

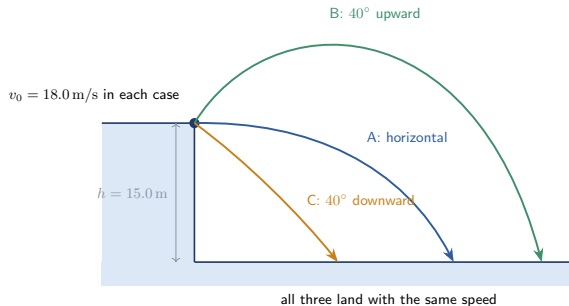
the work of some forces can be *stored* and given back.

Which forces, and how do we use this?

Opening question

Three identical balls leave the top of a 15.0 m cliff with the same speed, 18.0 m/s: A horizontally, B at 40° above the horizontal, C at 40° below it. Neglect air resistance. Which ball hits the ground with the greatest speed?

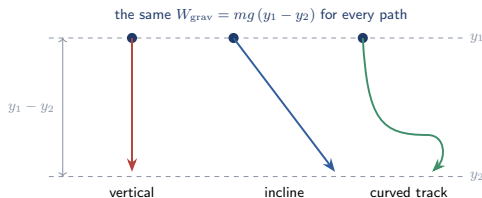
The three launches



Predict first

Sketch the three paths and mark the change in height of each ball. Then predict: A, B or C fastest, or all equal? (Hint: Week 4, work done by the weight.)

The work of gravity depends only on the height



Rule: work done by gravity (near the Earth's surface)

$$W_{\text{grav}} = mg(y_1 - y_2)$$

for **any** path from height y_1 to height y_2 . Only the change in height matters, not the path.

From Week 4: $W = \vec{F} \cdot \vec{s}$ with $\vec{F} = m\vec{g}$ downward. A horizontal step contributes nothing; the vertical steps add up to $y_1 - y_2$.

Defining gravitational potential energy

Rule: gravitational potential energy

$$W_{\text{grav}} = -\Delta U_{\text{grav}} \quad U_{\text{grav}} = mgy$$

y is the height above a reference level of your choice (y positive upward). Unit: joule.

Physical meaning

U is the work gravity *can do* as the object moves down: energy **stored** in the arrangement of the object and the Earth.

Common error

U_{grav} is shared by the object and the Earth; it is not a property of the object alone. It increases when the object moves *up*.

The zero of U is a choice

A 0.50 kg ball moves from A (2.0 m above the floor) to B (0.50 m above the floor).

Reference level ($y = 0$)	U_A	U_B	$\Delta U = U_B - U_A$
the floor	9.8 J	2.45 J	-7.35 J
1.0 m above the floor ($y_A = +1.0$, $y_B = -0.5$)	4.9 J	-2.45 J	-7.35 J

What is physical

U itself changes with the reference level, and can be negative. The change ΔU , and hence the work of gravity $W = -\Delta U = +7.35 \text{ J}$, is the same. Choose the reference level that makes the problem simplest, then keep it.

TOOLKIT 1 – Gravitational potential energy and the reference level

1. **SKETCH** the motion and mark the heights of the object at the states you need.
2. **REFERENCE:** choose $y = 0$ at the lowest point (or at the start), and write it on the sketch.
3. **HEIGHTS:** measure y upward from the reference; heights below are negative.
4. **CALCULATE** $U = mgy$ at each state, or $\Delta U = mg \Delta y$ directly.
5. **PREDICT** the sign of ΔU : positive when the object rises, negative when it falls.

RULE: only differences in U have physical meaning.

WATCH OUT: use the vertical change in height, not the distance along a slope; do not change the reference level in the middle of a problem.

CHECK: does ΔU equal $-W_{\text{grav}}$, with the correct sign?

Only gravity does work on the object. From Week 4 and the rule above:

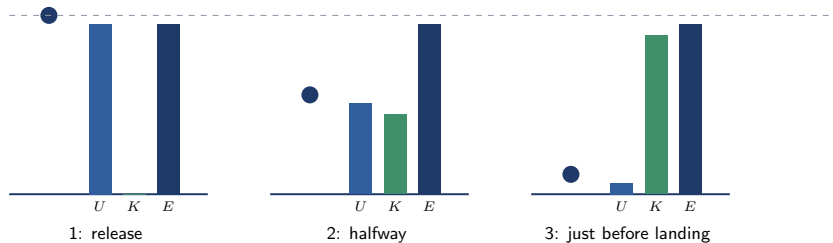
1. Work–energy theorem: $W_{\text{tot}} = \Delta K = K_2 - K_1$.
2. Only gravity does work: $W_{\text{tot}} = W_{\text{grav}} = -\Delta U = U_1 - U_2$.
3. Combine: $K_2 - K_1 = U_1 - U_2 \Rightarrow K_1 + U_1 = K_2 + U_2$.

Rule: conservation of mechanical energy

$$E = K + U = \text{constant} \quad \text{so} \quad K_1 + U_1 = K_2 + U_2$$

valid when **only conservative forces do work**. Other forces may act if they do no work (normal force, tension, perpendicular to the motion).

A bar-chart representation



Predict first

A ball falls from rest with no air resistance. Sketch the K and U bars at the halfway state before looking: what must their heights add to?

U falls and K rises by the same amount: energy *changes form*, but $E = K + U$ stays constant.
 K itself is not conserved.

TOOLKIT 2 – Conservation of mechanical energy

1. **STATES:** choose an initial point (1) and a final point (2); mark heights and speeds on the sketch.
2. **FORCES:** list every force on the object. Confirm that all except gravity (and later the spring) do *no* work.
3. **BARS:** sketch the K and U bars at each state; their total must be equal.
4. **WRITE** $\frac{1}{2}mv_1^2 + mgy_1 = \frac{1}{2}mv_2^2 + mgy_2$. The mass often cancels.
5. **SOLVE** for the unknown speed or height.

WATCH OUT: if a force that does work (friction, a push) is present, this equation fails; use Toolkit 4.

CHECK: is the speed larger where the object is lower? Does E come out the same at a third point?

Question

In the cliff problem ($h = 15.0 \text{ m}$, $v_0 = 18.0 \text{ m/s}$, no air resistance), find the speed of a ball just before it hits the ground when it is thrown (a) horizontally, (b) 40° above the horizontal, (c) 40° below the horizontal.

Predict first

Sketch the heights. Does the direction of the launch change the change in height? Roughly, is the final speed more or less than 18 m/s ? More or less than 30 m/s ?

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Predict first

Sketch the heights. Does the direction of the launch change the change in height? Roughly, is the final speed more or less than 18 m/s? More or less than 30 m/s?

Set up: $y = 0$ at the ground; $y_1 = 15.0$ m, $y_2 = 0$. Only gravity does work, so $K_1 + U_1 = K_2 + U_2$.

Execute:

$$\frac{1}{2}mv_0^2 + mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{v_0^2 + 2gh} = \sqrt{18.0^2 + 2(9.80)(15.0)} = \sqrt{618} = 24.9 \text{ m/s.}$$

Evaluate: the same answer for (a), (b) and (c): 24.9 m/s, between 18 and 30 m/s. Only $y_1 - y_2$ and v_0 matter, not the direction. ✓

The cliff ball: the highest point

Question

For ball B (launched 40° above the horizontal at 18.0 m/s), find the maximum height above the cliff top. What is its speed at that point?

Predict first

Is the speed at the highest point zero? What does the velocity consist of at that point?

The cliff ball: the highest point

Question

For ball B (launched 40° above the horizontal at 18.0 m/s), find the maximum height above the cliff top. What is its speed at that point?

Predict first

Is the speed at the highest point zero? What does the velocity consist of at that point?

Set up: state 1: the launch point ($y = 0$ at the cliff top); state 2: the highest point, where $v_y = 0$ but $v_x = v_0 \cos 40^\circ$ is unchanged (no horizontal force).

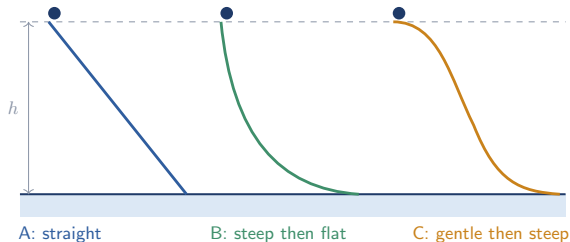
Execute:

$$\frac{1}{2}mv_0^2 = \frac{1}{2}m(v_0 \cos 40^\circ)^2 + mgH \Rightarrow H = \frac{v_0^2 (1 - \cos^2 40^\circ)}{2g} = \frac{(18.0)^2 \sin^2 40^\circ}{2(9.80)} = 6.83 \text{ m.}$$

Speed at the top: $v_0 \cos 40^\circ = 13.8 \text{ m/s}$, not zero.

Evaluate: this matches the Week 2 projectile result $H = (v_0 \sin \theta)^2 / 2g$. ✓

Concept check: ramps and reference levels

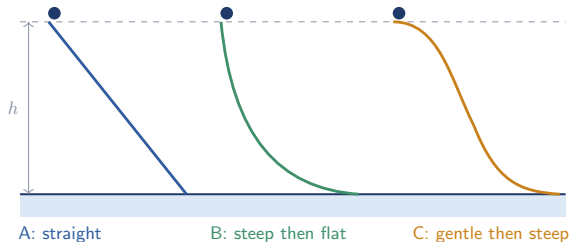


Three frictionless tracks start at the same height h and end at the same level. An object is released from rest at the top of each.

Answer, with a reason

(a) Compare the speeds at the bottom. (b) Do the arrival times have to be equal? (c) If the reference level were moved to the top, which of U_1 , U_2 , ΔU would change?

Concept check: ramps and reference levels



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Answer, with a reason

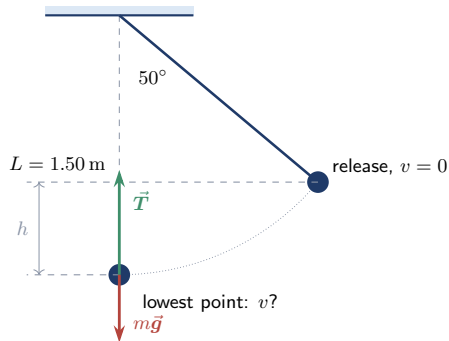
(a) Compare the speeds at the bottom. (b) Do the arrival times have to be equal? (c) If the reference level were moved to the top, which of U_1 , U_2 , ΔU would change?

(a) All equal: $v = \sqrt{2gh}$; the normal force does no work. (b) No: the acceleration along each track differs (Week 3), so the times differ. (c) U_1 and U_2 change; ΔU does not.

Curved paths: energy for the speed, Newton for the force

Question

A 0.80 kg bob hangs on a string of length 1.50 m. It is released from rest with the string 50° from the vertical. Find (a) its speed at the lowest point, and (b) the tension in the string there.



Predict first

Sketch the free-body diagram at the bottom. Is T greater than, equal to, or less than $mg = 7.84\text{ N}$?

The pendulum: solution

Set up: $y = 0$ at the lowest point. The tension is always perpendicular to the motion, so it does no work; only gravity does. Starting height:

$$h = L(1 - \cos 50^\circ) = 1.50(1 - 0.643) = 0.536 \text{ m}.$$

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$$h = L(1 - \cos 50^\circ) = 1.50(1 - 0.643) = 0.536 \text{ m.}$$

(a) Energy: $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2(9.80)(0.536)} = 3.24 \text{ m/s.}$

Set up: $y = 0$ at the lowest point. The tension is always perpendicular to the motion, so it does no work; only gravity does. Starting height:

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(a) Energy: $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2(9.80)(0.536)} = 3.24 \text{ m/s.}$

(b) Newton's second law (energy cannot give a force): circular motion, centre upward at the bottom:

$$T - mg = \frac{mv^2}{L} \Rightarrow T = mg + \frac{mv^2}{L} = 7.84 + \frac{(0.80)(10.5)}{1.50} = 7.84 + 5.60 = 13.4 \text{ N.}$$

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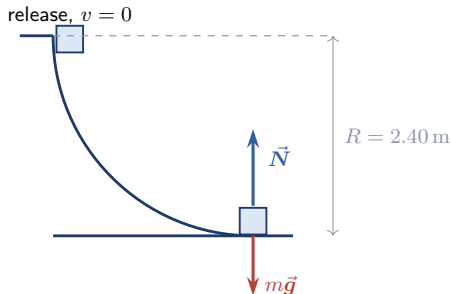
Evaluate: $T > mg$, as predicted: the bob accelerates upward. The energy method gave v without the path shape; Newton's law then gave the force at one point. Check:

$$T = mg(3 - 2 \cos 50^\circ) = 13.4 \text{ N. } \checkmark$$

Your turn: a block in a smooth bowl

Your turn

A 1.50 kg block is released from rest at the rim of a smooth quarter-circular bowl of radius $R = 2.40\text{ m}$. Find (a) its speed at the bottom, and (b) the normal force of the bowl on the block there.



Predict first

Sketch the free-body diagram at the bottom. Is N larger than $mg = 14.7\text{ N}$? By roughly what factor?

Set up: $y = 0$ at the bottom; $y_1 = R$. The normal force is perpendicular to the motion and does no work.

Set up: $y = 0$ at the bottom; $y_1 = R$. The normal force is perpendicular to the motion and does no work.

(a) $mgR = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gR} = \sqrt{2(9.80)(2.40)} = 6.86 \text{ m/s}.$

Set up: $y = 0$ at the bottom; $y_1 = R$. The normal force is perpendicular to the motion and does no work.

(a) $mgR = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gR} = \sqrt{2(9.80)(2.40)} = 6.86 \text{ m/s}.$

(b) $N - mg = \frac{mv^2}{R} \Rightarrow N = mg + \frac{m(2gR)}{R} = 3mg = 44.1 \text{ N}.$

Set up: $y = 0$ at the bottom; $y_1 = R$. The normal force is perpendicular to the motion and does no work.

(a) $mgR = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gR} = \sqrt{2(9.80)(2.40)} = 6.86 \text{ m/s}.$

(b) $N - mg = \frac{mv^2}{R} \Rightarrow N = mg + \frac{m(2gR)}{R} = 3mg = 44.1 \text{ N}.$

Evaluate: $N = 3mg$ for *any* radius: three times the weight, larger than mg as predicted. The mass cancels in (a); the radius cancels in (b). ✓

Contact 1: what we can now do

Work of gravity (path independent)



Potential energy $U = mgy$, $W = -\Delta U$



Conservation $K_1 + U_1 = K_2 + U_2$

You should be able to

- calculate U_{grav} with a stated reference level;
- explain why only ΔU matters;
- state when $E = K + U$ is conserved;
- find speeds and heights on straight and curved paths, using Newton's law for forces.

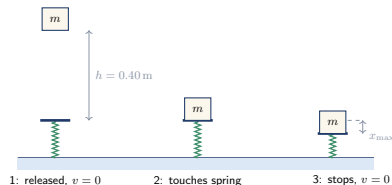
✓ Before Contact 2

A real ball dropped on the floor does not return to its starting height. Where did the energy go? Read Young and Freedman, §7.2.

Contact 2: Quantify

How do springs, friction and other forces fit in?

An opening question: a block dropped on a spring



Poll

A block is dropped from 0.40 m above a light vertical spring. At state 3 it momentarily stops. Is state 3 the position where the spring force equals the weight (the equilibrium point)?

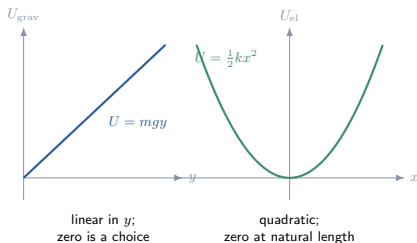
Predict first

Sketch the velocity of the block against height. Is it moving fastest at the equilibrium point, or at the instant it touches the spring?

Rule: elastic potential energy of a spring

$$W_{\text{el}} = -\Delta U_{\text{el}} \quad U_{\text{el}} = \frac{1}{2}kx^2$$

x is the stretch or compression from the **natural length**. $U_{\text{el}} \geq 0$ for both stretch and compression.



Physical meaning

The work a spring *can do* as it returns to natural length. Week 4: the spring's work from x_1 to x_2 is $\frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2 = -\Delta U_{\text{el}}$.

TOOLKIT 3 – Elastic potential energy and springs

1. NATURAL LENGTH: mark it on the sketch; measure x from it (stretch or compression).

2. STATES: give x at each state, with $x = 0$ where the spring is unstretched.

3. CALCULATE $U_{\text{el}} = \frac{1}{2}kx^2$; for a vertical spring add $U_{\text{grav}} = mgy$ with one reference level.

4. COMBINE: $E = K + U_{\text{grav}} + U_{\text{el}}$ (Toolkit 2 with an extra term).

RULE: Hooke's law holds for small stretches only; check that x is within the elastic range.

WATCH OUT: use the stretch from natural length, not the spring's length; the equilibrium position of a hanging mass is $x = mg/k$, which is not where the mass stops after a drop.

CHECK: is k in N/m, x in m, and U_{el} never negative?

Question

A 2.00 kg block is released from rest 0.40 m above a light vertical spring ($k = 1500 \text{ N/m}$). Find the maximum compression x_{max} of the spring.

Predict first

Sketch the three states. Neglect the compression of the spring in the fall: estimate x_{max} from $\frac{1}{2}kx^2 = mgh$. Will the true value be larger or smaller than your estimate? Compare with the equilibrium compression mg/k .

Dropping a block on a spring: solution

Set up: $y = 0$ at the maximum compression; state 1: released ($v = 0$, height $h + x_{\max}$, spring natural); state 2: maximum compression ($v = 0$). Gravity and the spring are the only forces doing work.

Execute: $mg(h + x_{\max}) = \frac{1}{2}kx_{\max}^2 \Rightarrow 750x^2 - 19.6x - 7.84 = 0$ (in newtons, metres),

$$x_{\max} = \frac{19.6 + \sqrt{19.6^2 + 4(750)(7.84)}}{2(750)} = \frac{19.6 + 154.6}{1500} = 0.116 \text{ m.}$$

Evaluate: estimate $\sqrt{2mgh/k} = 0.102 \text{ m}$, slightly smaller, as predicted. The equilibrium compression is only $mg/k = 0.013 \text{ m}$: the block overshoots it by far. ✓

When other forces act

Gravity and the spring have potential energies. A friction force, a push, or air resistance does not. Their work enters separately:

Rule: energy equation with other forces

$$K_1 + U_1 + W_{\text{other}} = K_2 + U_2$$

$U = U_{\text{grav}} + U_{\text{el}}$. W_{other} is the work done by all forces **except** gravity and the spring; equivalently $W_{\text{other}} = \Delta E = \Delta(K + U)$.

- $W_{\text{other}} > 0$: a force adds mechanical energy (a push, a motor).
- $W_{\text{other}} < 0$: a force removes mechanical energy (friction, drag).

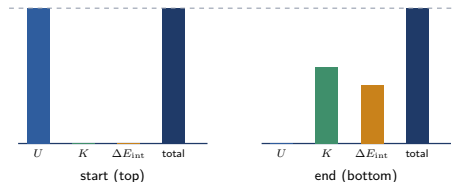
The system

The system is the object, the Earth and the spring. Forces inside the system with a potential energy are in U . All other forces are outside, and do work on the system: that is W_{other} .

Rule: kinetic friction

$$W_f = -f_k d \quad \Delta E_{\text{int}} = +f_k d$$

d is the path length. The mechanical energy lost appears as **thermal (internal) energy** of the surfaces, which warm up.



Energy is not destroyed

Friction does negative work, but the energy changes form: mechanical to thermal. Total energy is conserved (see the law later this week); only the *mechanical* energy decreases. Temperature is treated in the thermal physics part.

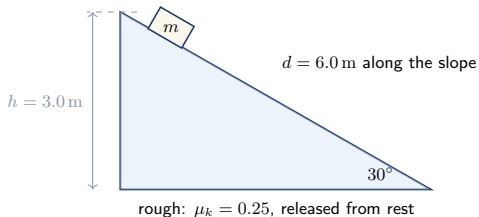
TOOLKIT 4 – Energy equation with other forces

1. **SYSTEM:** object, Earth, spring. List the forces that are *not* gravity or the spring.
 2. **WORK OF EACH OTHER FORCE:** sign first (friction < 0), then $W = \vec{F} \cdot \vec{s}$ (Week 4, Toolkits 1 and 2). Forces perpendicular to the motion contribute zero.
 3. **STATES AND BARS:** K , U_{grav} , U_{el} at 1 and 2, with a thermal-energy bar if friction acts.
 4. **WRITE** $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$ and solve.
 5. **CHOOSE THE METHOD:** energy for speed, height or distance between two states; Newton's second law for acceleration, time, or a force at a point.
- WATCH OUT:** friction uses the *path length*, not the displacement; W_{other} does not include the work of gravity or the spring, which is already in U .
- CHECK:** is the final speed smaller than in the frictionless case? Does the second method agree?

Lecturer example: a box on a rough incline

Question

A 5.00 kg box is released from rest on a 30° incline and slides 6.00 m along it. The coefficient of kinetic friction is 0.25. Find its speed after the 6.00 m.



Predict first

Sketch the free-body diagram. Is the speed more or less than the frictionless value $\sqrt{2gh}$? Which force does negative work?

The incline: solution

Set up: $y = 0$ at the bottom. The drop in height is $h = 6.00 \sin 30^\circ = 3.00 \text{ m}$.
 $n = mg \cos 30^\circ = 42.4 \text{ N}$, $f_k = \mu_k n = 10.6 \text{ N}$. The normal force does no work.

The incline: solution

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 $n = mg \cos 30^\circ = 42.4 \text{ N}$, $f_k = \mu_k n = 10.6 \text{ N}$. The normal force does no work.

Execute: $K_1 + U_1 + W_f = K_2 + U_2$: $0 + mgh - f_k d = \frac{1}{2}mv^2$
 $147 - (10.6)(6.00) = 147 - 63.7 = 83.3 \text{ J} = \frac{1}{2}(5.00)v^2 \Rightarrow v = 5.77 \text{ m/s}$.

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 $147 - (10.6)(6.00) = 147 - 63.7 = 83.3 \text{ J} = \frac{1}{2}(5.00)v^2 \Rightarrow v = 5.77 \text{ m/s}$.

Evaluate: the frictionless speed is $\sqrt{2gh} = 7.67 \text{ m/s}$, so the answer is smaller, as predicted;
63.7 J became thermal energy. Newton's second law:
 $a = g(\sin 30^\circ - 0.25 \cos 30^\circ) = 2.78 \text{ m/s}^2$ and $v = \sqrt{2(2.78)(6.00)} = 5.77 \text{ m/s}$. ✓

Physical meaning

Energy gave v in one step with no acceleration; Newton's law also gives the time. Choose by what is asked (Week 4, Toolkit 4).

Concept check: which forces go into U ?

For each force: potential energy term, or in W_{other} ? Give the sign if it is in W_{other} .

1. The weight of a block sliding down a rough incline.
2. The force of a spring that pushes the block up the incline.
3. Kinetic friction between the block and the incline.
4. The normal force of the incline on the block.
5. A person pushing the block up the incline at constant speed.

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4. The normal force of the incline on the block.
5. A person pushing the block up the incline at constant speed.

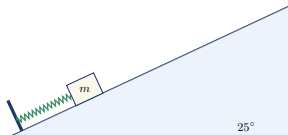
(1) U_{grav} . (2) U_{el} . (3) $W_{\text{other}} < 0$. (4) $W_{\text{other}} = 0$ (perpendicular to the motion). (5) $W_{\text{other}} > 0$.

Common error: counting the weight twice, once in U_{grav} and once as a work term.

Your turn: a spring fires a block up a rough incline

Your turn

A 0.60 kg block is pressed against a spring ($k = 600 \text{ N/m}$) at the bottom of a 25° incline, compressing it 0.15 m along the incline, and released from rest. The block is not attached; $\mu_k = 0.20$. (a) How far along the incline from the release point does it travel before stopping? (b) Does it then remain at rest?



Predict first

Which terms are in U , and which in W_{other} ? Is the distance more or less than 2 m? Compare $\tan 25^\circ$ with μ_k before you answer (b).

Set up: state 1: compressed, $v = 0$, $U_{\text{el}} = \frac{1}{2}(600)(0.15)^2 = 6.75 \text{ J}$; state 2: stopped at distance s along the incline, $U_{\text{el}} = 0$. Height gained $s \sin 25^\circ$; friction $f_k = \mu_k mg \cos 25^\circ = 1.07 \text{ N}$.

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Execute: $K_1 + U_1 + W_f = K_2 + U_2$: $6.75 - f_k s = mgs \sin 25^\circ$
 $6.75 = s(mg \sin 25^\circ + f_k) = s(2.485 + 1.066) \Rightarrow s = 1.90 \text{ m}$.

Set up: state 1: compressed, $v = 0$, $U_{\text{el}} = \frac{1}{2}(600)(0.15)^2 = 6.75 \text{ J}$; state 2: stopped at distance s along the incline, $U_{\text{el}} = 0$. Height gained $s \sin 25^\circ$; friction $f_k = \mu_k mg \cos 25^\circ = 1.07 \text{ N}$.

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 $6.75 = s(mg \sin 25^\circ + f_k) = s(2.485 + 1.066) \Rightarrow s = 1.90 \text{ m}$.

(b) To stay at rest the friction needed is $mg \sin 25^\circ = 2.49 \text{ N}$. Even if the static coefficient equals the kinetic one, friction can supply at most $\mu mg \cos 25^\circ = 1.07 \text{ N}$ (equivalently $\tan 25^\circ = 0.47 > 0.20$). So the block does **not** remain at rest: it slides back down.

Evaluate: $s = 1.90 \text{ m} < 2 \text{ m}$; the block leaves the spring at $v = 4.55 \text{ m/s}$ after 0.15 m and then decelerates over 1.75 m under gravity and friction. ✓

Contact 2: what we can now do

Spring $U_{\text{el}} = \frac{1}{2}kx^2$



Gravity and spring together

$$E = K + U_{\text{grav}} + U_{\text{el}}$$



Other forces $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$

You should be able to

- calculate U_{el} from the natural length;
- combine U_{grav} and U_{el} in one equation;
- include friction and other forces, and say where the lost energy goes;
- choose between energy and Newton's second law.

✓ Before Contact 3

Gravity and springs can be stored. Friction cannot. What is the test? Read Young and Freedman, §7.3–7.5.

Contact 3: Extend

Which forces have a potential energy, and what does $U(x)$ tell us?

Which forces can be stored?

Opening question

A crate is moved from A to B by two different paths. Gravity does the same work on both. Does friction? Does every force have a potential energy? How would we test?

Predict first

Recall the cliff problem and the incline: for which forces did the path not matter?

Rule: conservative force

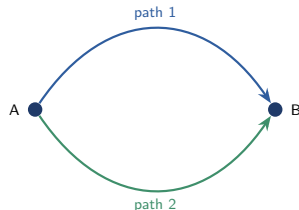
The work of a conservative force

1. can be written as $W = -\Delta U$;
2. is reversible;
3. is independent of the path between two points;
4. is zero for any closed path.

Examples: gravity, the spring force.

Otherwise the force is

nonconservative.



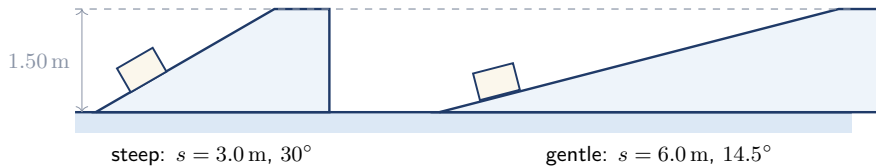
conservative: $W_1 = W_2$, and the work around the loop $A \rightarrow B \rightarrow A$ is zero

Nonconservative: kinetic friction, drag, a push or pull by a person, an explosion. Some dissipate mechanical energy (friction); some add it (a motor).

TOOLKIT 5 – Classifying forces

1. **TWO PATHS:** pick two paths between the same two points; compute the work of the force on each.
 2. **COMPARE:** equal work on every path means conservative; different work means nonconservative.
 3. **LOOP:** equivalently, the work around a closed path is zero (conservative) or not (nonconservative).
 4. **ASSIGN:** conservative forces go into U ; nonconservative forces go into W_{other} .
- WATCH OUT:** the work of friction depends on the path length, so it cannot be written as $-\Delta U$; forces that do no work (normal force, tension) simply do not appear in the energy equation.
- CHECK:** does the work of friction depend on the path length? Is it always negative for a sliding object?

Lecturer example: two ramps



Question

A 10.0 kg crate is pushed at constant speed up either a steep ramp (3.0 m long) or a gentle ramp (6.0 m long) to a platform 1.50 m high, with the push parallel to the ramp and $\mu_k = 0.20$. Find the work of gravity, the work of friction and the work you do on each ramp.

Predict first

Which ramp needs more force? Which needs more work? Is the work of gravity the same?

Two ramps: solution

Gravity (conservative): $W_{\text{grav}} = -mgh = -(10.0)(9.80)(1.50) = -147 \text{ J}$ on both ramps (path independent).

Two ramps: solution

Gravity (conservative): $W_{\text{grav}} = -mgh = -(10.0)(9.80)(1.50) = -147 \text{ J}$ on both ramps (path independent).

Friction (nonconservative), steep: $\cos \theta = 0.866$, $f_k = 0.20(98.0)(0.866) = 17.0 \text{ N}$,
 $W_f = -(17.0)(3.0) = -50.9 \text{ J}$.

Friction, gentle: $\sin \theta = 0.25$, $\cos \theta = 0.968$, $f_k = 19.0 \text{ N}$, $W_f = -(19.0)(6.0) = -113.9 \text{ J}$.

Two ramps: solution

Gravity (conservative): $W_{\text{grav}} = -mgh = -(10.0)(9.80)(1.50) = -147 \text{ J}$ on both ramps (path independent).

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Your work ($\Delta K = 0$, so $W_{\text{you}} + W_{\text{grav}} + W_f = 0$): steep $147 + 50.9 = 198 \text{ J}$; gentle
 $147 + 113.9 = 261 \text{ J}$.

Two ramps: solution

Gravity (conservative): $W_{\text{grav}} = -mgh = -(10.0)(9.80)(1.50) = -147 \text{ J}$ on both ramps (path independent).

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Your work ($\Delta K = 0$, so $W_{\text{you}} + W_{\text{grav}} + W_f = 0$): steep $147 + 50.9 = 198 \text{ J}$; gentle
 $147 + 113.9 = 261 \text{ J}$.

Evaluate: the gentle ramp needs a smaller force ($F = mg \sin \theta + f_k = 43.5 \text{ N}$ against 66.0 N) but more *work*, because friction acts over a longer path. Check: $(66.0)(3.0) = 198 \text{ J}$ and $(43.5)(6.0) = 261 \text{ J}$. ✓

The law of conservation of energy

Rule: law of conservation of energy

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0 \quad (\text{isolated system})$$

Energy is never created or destroyed. It is **transformed** between forms (kinetic, potential, thermal) or **transferred** across the boundary of the system as work W_{other} .

Concept check

Identify the transformation: (a) a car brakes to rest; (b) a person climbs a staircase at constant speed.

- (a) $K \rightarrow E_{\text{int}}$ (brakes and tyres warm up).
- (b) chemical energy in the person $\rightarrow U_{\text{grav}}$ and E_{int} (the muscles warm up).

Rule: force from potential energy (one dimension)

$$F_x = -\frac{dU}{dx}$$

The force is minus the slope of $U(x)$: it points toward **lower** potential energy.

Check against known forces:

Spring: $U = \frac{1}{2}kx^2 \Rightarrow F_x = -kx$ (Hooke's law).

Gravity: $U = mgy \Rightarrow F_y = -mg$ (downward).

Physical meaning

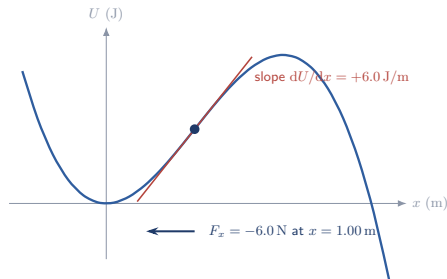
Because $W = -\Delta U$, a small step dx gives $F_x dx = -dU$. The force is the rate at which stored energy decreases with position.

TOOLKIT 6 – Force from potential energy

1. **SKETCH** $U(x)$, or read it from a graph.
 2. **SLOPE:** find dU/dx at the point (derivative of the formula, or slope of the tangent).
 3. **FORCE:** $F_x = -dU/dx$. Predict the direction first: toward lower U .
 4. **EQUILIBRIUM:** $F_x = 0$ where the slope is zero.
- WATCH OUT:** the sign of U does not give the direction of the force; the sign of the *slope* does.
- CHECK:** does the force point down the slope? Do the units work out ($\text{J/m} = \text{N}$)?

Question

A particle moves along x with potential energy $U(x) = 6.0x^2 - 2.0x^3$ (joules, x in metres). Find (a) the force at $x = 1.0$ m, and (b) the equilibrium positions.



Predict first

Look at the slope of U at $x = 1.0$ m. Does the force point toward $+x$ or $-x$?

(a) Execute: $F_x = -\frac{dU}{dx} = -(12x - 6x^2) = -12x + 6x^2$; at $x = 1.0$ m:
 $F_x = -12 + 6 = -6.0$ N.

(a) Execute: $F_x = -\frac{dU}{dx} = -(12x - 6x^2) = -12x + 6x^2$; at $x = 1.0$ m:
 $F_x = -12 + 6 = -6.0$ N.

(b) Equilibrium: $F_x = 0 \Rightarrow 6x(x - 2) = 0 \Rightarrow x = 0$ and $x = 2.0$ m, where $U = 0$ and $U = 8.0$ J.

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(b) Equilibrium: $F_x = 0 \Rightarrow 6x(x - 2) = 0 \Rightarrow x = 0$ and $x = 2.0$ m, where $U = 0$ and $U = 8.0$ J.

Evaluate: (a) the slope at $x = 1.0$ m is $+6.0$ J/m, so the force is -6.0 N, toward lower U , as predicted. ✓

At $x = 0$, U is a minimum: a small displacement gives a force back toward $x = 0$ (**stable**). At $x = 2.0$ m, U is a maximum: a small displacement gives a force away (**unstable**).

Rule: energy diagram

Plot $U(x)$ and the constant total energy E on one graph. At each x :

$$K = E - U \geq 0$$

- **Allowed** region: $E \geq U$. **Forbidden**: $E < U$ ($K < 0$ is impossible).
- **Turning points**: $E = U$, so $K = 0$ and $v = 0$; the motion reverses.
- **Equilibrium**: slope of U is zero (minimum: stable; maximum: unstable).
- If E is above every barrier, the particle can **escape**.

Physical meaning

The graph of U is the “landscape”; E is the level of the water. The particle can only be where the landscape is below the water.

TOOLKIT 7 – Reading energy diagrams

1. **DRAW** $U(x)$ and a horizontal line at E on the same axes.
2. **ALLOWED REGION:** shade where $E \geq U$; the particle is confined to the connected region it starts in.
3. **TURNING POINTS:** solve $U(x) = E$ (equal values of E and U).
4. **SPEED:** $K = E - U$, so $v = \sqrt{2(E - U)/m}$; fastest where U is lowest.
5. **EQUILIBRIUM:** minima are stable, maxima unstable.

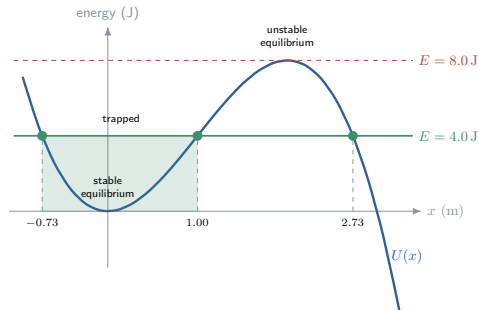
WATCH OUT: a particle cannot cross a region where $U > E$; the region it starts in matters, even if another allowed region exists elsewhere.

CHECK: is $E - U \geq 0$ at every point you claim is reachable?

Lecturer example: trapped or not?

Question

For $U(x) = 6.0x^2 - 2.0x^3$ (joules), a 0.50 kg particle at $x = 0$ has total energy $E = 4.0$ J. Find (a) the turning points of its motion, (b) its speed at $x = 0$, and (c) the energy above which it can escape to large x .



Predict first

Sketch your own $U(x)$ and the line $E = 4.0$ J, then compare with the diagram. Which way can the particle go before it turns? Is it fastest at $x = 0$?

(a) Turning points: $6x^2 - 2x^3 = 4 \Rightarrow x^3 - 3x^2 + 2 = 0 = (x - 1)(x^2 - 2x - 2)$, so $x = 1.00$, $1 \pm \sqrt{3} = 2.73, -0.73$ m. The particle starts at $x = 0$, and $U < E$ between -0.73 and 1.00 , so it oscillates between $x = -0.73$ m and $x = +1.00$ m.

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(b) Speed at $x = 0$: $K = E - U = 4.0 - 0 = 4.0 \text{ J} = \frac{1}{2}(0.50)v^2 \Rightarrow v = 4.0 \text{ m/s}$.

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(b) Speed at $x = 0$: $K = E - U = 4.0 - 0 = 4.0 \text{ J} = \frac{1}{2}(0.50)v^2 \Rightarrow v = 4.0 \text{ m/s}$.

(c) Escape: the barrier top is $U(2.0) = 8.0 \text{ J}$. For $E > 8.0 \text{ J}$ the particle can pass $x = 2.0$ m and escape.

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(c) Escape: the barrier top is $U(2.0) = 8.0 \text{ J}$. For $E > 8.0 \text{ J}$ the particle can pass $x = 2.0$ m and escape.

Evaluate: at $x = 0$ the speed is the maximum, since U is lowest there (stable equilibrium). The region $x > 2.73$ m is also allowed for $E = 4.0 \text{ J}$, but it is separated from the starting region by $U > E$: the particle cannot reach it. ✓

Your turn

A 0.25 kg particle has potential energy $U(x) = 0.50x^4 - 2.0x^2$ (joules, x in metres). (a) Find the force at $x = 1.0$ m. (b) Find the equilibrium positions and say which are stable. (c) The total energy is $E = -1.0$ J and the particle is in the right-hand well. Find its turning points and its greatest speed.

Predict first

Sketch $U(x)$: how many minima does it have? Where is the particle fastest, and is U negative there?

A double well: solution

(a) $F_x = -(2x^3 - 4x) = 4x - 2x^3$; at $x = 1.0$: $+2.0$ N (toward $+x$, down the slope toward the minimum at $x = \sqrt{2}$).

(b) $F = 0 \Rightarrow x = 0, \pm\sqrt{2} = \pm 1.41$ m. $U(0) = 0$ is a maximum (unstable); $U(\pm 1.41) = -2.0$ J are minima (stable).

(c) $U = E \Rightarrow x^4 - 4x^2 + 2 = 0 \Rightarrow x^2 = 2 \pm \sqrt{2}$: turning points $x = 0.77$ m and 1.85 m. At $x = 1.41$ m: $K = -1.0 - (-2.0) = 1.0$ J, so $v = \sqrt{2(1.0)/0.25} = 2.8$ m/s.

Integrated problem: a spring, a rough strip and a smooth curve

Decide what physics applies; do not look for a formula

A 0.40 kg block is pushed against a horizontal spring ($k = 1200 \text{ N/m}$), compressing it 0.080 m, and released from rest. The block is not attached. The track is smooth except for a 1.2 m rough strip ($\mu_k = 0.35$), followed by a smooth curve.

- (a) How high up the curve does the block rise?
- (b) The block slides back down. By how much does it compress the spring when it returns?



Predict first

Sketch the energy bars. Does the block return with less energy than it started with?

Identify, set up: $U_{\text{el}} = \frac{1}{2}kx^2 = \frac{1}{2}(1200)(0.080)^2 = 3.84 \text{ J}$. Friction acts only on the strip:
 $f_k = \mu_k mg = 0.35(0.40)(9.8) = 1.372 \text{ N}$, $W_f = -f_k(1.2) = -1.65 \text{ J}$ on each crossing. The curve is smooth.

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(a) Release $\rightarrow$ highest point ($K = 0$):

$$U_{\text{el},1} + W_f = mgH \Rightarrow 3.84 - 1.65 = 2.19 = (0.40)(9.8)H \Rightarrow H = 0.56 \text{ m}.$$

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(b) Release $\rightarrow$ back at the spring (second crossing, $K = 0$ at maximum compression x):

$$3.84 - 2(1.65) = 0.55 \text{ J} = \frac{1}{2}kx^2 \Rightarrow x = \sqrt{\frac{2(0.547)}{1200}} = 0.030 \text{ m}.$$

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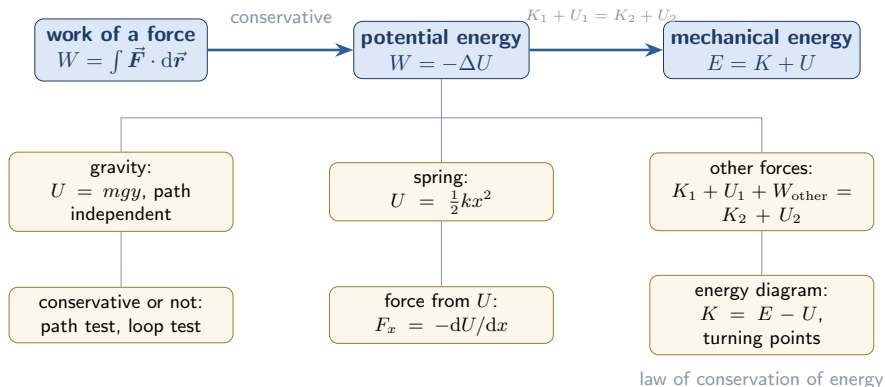
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Evaluate: the compression is 3.0 cm, less than 8.0 cm, as predicted: each crossing removes 1.65 J of mechanical energy, which becomes thermal energy. E_{int} after the round trip is 3.29 J, and $3.29 + 0.55 = 3.84 \text{ J}$. ✓

Week 5 concept map



MASTER PROCEDURE – ENERGY

- 1. IDENTIFY** The object, and what is asked: a speed, a height, a distance, a force or a compression.
 - 2. SKETCH AND STATES** Mark the two states with heights, speeds and spring extensions; choose the reference level.
 - 3. SYSTEM AND FORCES** Object, Earth, spring. Which forces are in U ; which are in W_{other} ; which do no work.
 - 4. PREDICT** Does mechanical energy stay the same, or decrease or increase? Rough size of the answer.
 - 5. SELECT THE METHOD** Conservation of E , the energy equation with W_{other} , or Newton's second law (a force at a point, time).
 - 6. EXECUTE** $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$ (Toolkits 1 to 4); or $F_x = -dU/dx$ and $K = E - U$ (Toolkits 6 and 7).
 - 7. EVALUATE** Units, sign, speed less than the frictionless value if friction acts, check by a second method.
- SKETCH → PREDICT → CALCULATE → CHECK.

Inside the course framework

IDENTIFY ↔ steps 1–2
SET UP ↔ steps 3–5
EXECUTE ↔ step 6
EVALUATE ↔ step 7

Physical meaning

Next lecture: forces that act for a short time, impulse and momentum, and their conservation in collisions.



THE SEVEN TOOLKITS

- | | | |
|---|---|---|
| 1 | Gravitational potential energy | $U = mgy$; reference level; only ΔU matters |
| 2 | Conservation of mechanical energy | two states; $K_1 + U_1 = K_2 + U_2$; bar charts |
| 3 | Elastic potential energy and springs | $U = \frac{1}{2}kx^2$ from natural length; combined with gravity |
| 4 | Energy equation with other forces | $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$; friction as thermal energy |
| 5 | Classifying forces | two-path test; closed loop; what goes into U |
| 6 | Force from potential energy | $F_x = -dU/dx$; slope; equilibrium |
| 7 | Reading energy diagrams | $K = E - U$; turning points; stability |

The Master Procedure sits above these. Week 1 to 4 toolkits (components, free-body diagrams, work, the work–energy theorem, springs) are reused, not renamed.

Potential energies

$$U_{\text{grav}} = mgy, \quad U_{\text{el}} = \frac{1}{2}kx^2$$

Work and potential energy

$$W_{\text{cons}} = -\Delta U$$

Conservation of mechanical energy

$$K_1 + U_1 = K_2 + U_2 \text{ (conservative forces only)}$$

With other forces

$$K_1 + U_1 + W_{\text{other}} = K_2 + U_2$$

Friction

$$W_f = -f_k d, \quad \Delta E_{\text{int}} = f_k d$$

Law of conservation of energy

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$

Force from potential energy

$$F_x = -dU/dx$$

Energy diagram

$$K = E - U \geq 0$$

Six habits for Week 5

✓ Habit 1

Sketch the states and choose the reference level first.

✓ Habit 2

List the forces: which are in U , which are in W_{other} , which do no work.

✓ Habit 3

Use the height change, not the path length, for U_{grav} ; use the path length for friction.

✓ Habit 4

Measure the spring stretch from its natural length.

✓ Habit 5

Energy for the speed; Newton's law for a force at a point.

✓ Habit 6

Check the sign, the units and a second method, and compare with your prediction.

Checklist: can I ...

- ☐ calculate $U_{\text{grav}} = mgy$ with a stated reference level?
- ☐ explain why only ΔU matters?
- ☐ state when mechanical energy is conserved?
- ☐ use $K_1 + U_1 = K_2 + U_2$?
- ☐ combine energy with Newton's second law on a curved path?
- ☐ calculate $U_{\text{el}} = \frac{1}{2}kx^2$?
- ☐ use the energy equation with friction and other forces?
- ☐ say where the lost mechanical energy goes?
- ☐ test whether a force is conservative?
- ☐ state the law of conservation of energy?
- ☐ find $F_x = -dU/dx$ and read an energy diagram?
- ☐ select the appropriate model in an unfamiliar problem?

After this week

Tutorial sheet: extra gravitational, spring, friction, force-from-potential and energy-diagram problems. Supplementary notes: the derivative refresher, the closed-loop proof, the derivation of $F = -dU/dx$, the vertical loop, stability, and the preview of $U = -GMm/r$.